

Problem Solving

1. Let $X_1, \dots, X_n$ be a random sample from a $N(0, \theta)$ distribution. We want to estimate the standard deviation $\sqrt{\theta}$. Find the constant c so that $Y = c \sum_{i=1}^n |X_i|$ is an unbiased estimator of $\sqrt{\theta}$ and determine its efficiency.

2. Let S^2 be the sample variance of a random sample of size $n > 1$ from $N(\mu, \theta)$, $0 < \theta < \infty$. where μ is known. We know $E(S^2) = \theta$.

(a) What is the efficiency of S^2 ?

(b) What is the MLE of θ ?

(c) What is the asymptotic distribution of $\sqrt{n}(\hat{\theta} - \theta)$?

